



**CALICUT UNIVERSITY – FOUR-YEAR UNDER
GRADUATE PROGRAMME (CU-FYUGP)**

BSc CHEMISTRY

Programme	B. Sc. Chemistry				
Course Title	PLASTICS AND FIBER TECHNOLOGY				
Type of Course	VOCATIONAL MINOR				
Semester	III				
Academic Level	200-299				
Course Details	Credit	Lecture per week	Tutorial per week	Practical per week	Total Hours
	4	3	-	2	75
Pre-requisites	Foundation course in polymer chemistry				
Course Summary	To impart the basic concepts of Mixing and compounding various moulding techniques. Understand about reinforced plastics and fiber technology				

Course Outcomes (CO):

CO	CO Statement	Cognitive Level*	Knowledge Category#	Evaluation Tools used
CO1	Understand the basics of plastic processing and the role of additives in plastics and Describe the principles and techniques involved in mixing and compounding of plastics	U	C	Class test /Assignment / Quiz
CO2	Explain and describe the principles and processes involved in plastic injection molding	R,U	C	Class test /Assignment / Quiz
CO3	Analyze the principles of calendering, laminating, and 3D printing in plastic processing	An	C	Class test /Assignment / Quiz
CO4	Understand the principles, processes involved and equipment used in	U	C	Class test /Assignment / Quiz

	compression molding, transfer molding, blow molding, rotational molding, and reaction injection molding			
CO5	Identify different types of fibers used in industries including cellulose derivatives, polyolefinic, polyester, polyamide, aramid, carbon, and glass fibers.	A	C	Class test /Assignment / Quiz
CO6	Describe fiber spinning operations and the manufacturing process of various types of cords used in the tire industry	U	C	Class test /Assignment / Quiz
CO7	Develop practical skill in different compounding , dipping and molding techniques	Ap	P	viva/ practical skill
* - Remember (R), Understand (U), Apply (Ap), Analyse (An), Evaluate (E), Create (C) # - Factual Knowledge(F) Conceptual Knowledge (C) Procedural Knowledge (P) Metacognitive Knowledge (M)				

Detailed Syllabus:

Module	Unit	Content	Hrs	Marks
I	Mixing and compounding		9	25
	1	Introduction to plastic processing, additives for plastics – Fillers, Antioxidants	1	
	2	Stabilisers, Colourants, Flame retardants	2	
	3	Plasticisers. Mixing and compounding of plastics	2	
	4	Mixing and compounding equipments- Compounding by batch mixer	2	
	5	High speed mixer - Two roll mill - Banbury Mixer - Ribbon blender - Planetary mixers	2	
II	Moulding techniques		18	25
	6	Plastic injection moulding, different types of injection moulding machines,	1	
	7	details of injection moulding machine, injection moulding of thermosets.	2	

	8	Extrusion, details of extruders, twin screw extruders, dies, post extrusion processing	2	
	9	Calendering. Laminating, 3D printing.	4	
	10	Compression moulding: hydraulic presses, press capacity and pressure calculations, moulding process.	1	
	11	Transfer moulding: moulding process and advantages, Blow moulding: extrusion and injection blow moulding.	4	
	12	Rotational moulding: process and equipment.	2	
	13	Reaction injection moulding: introduction, process and advantages.	2	
III	Reinforced plastics		9	24
	14	Reinforced plastics	1	
	15	Processing techniques	2	
	16	Hand lay-up, spray lay up	2	
	17	Filament winding autoclave, Bag moulding	4	
IV	Fiber technology		9	24
	18	Fibers from cellulose and its derivatives	1	
	19	Polyolefinic, polyester, polyamide, aramide	2	
	20	Carbon and glass fibers , Fiber spinning operations	2	
	21	Different types of cords used in tyre industry, definition of denier, tex, tenacity	2	
	22	Different types of twisting, geo textiles	2	
V		PRACTICALS	30	
		Any Five experiments from the following can be done 1. Preparation of dispersions of solid latex compounding 2. preparation of emulsions of liquid compounding ingredient 3. preparation of latex compounding for household gloves and finger caps 4. production of finger caps by dipping process 5. production of balloon by dipping process 6. practice production of table mat 7. Hands on training in production of injection moulded plastic articles 8. Hands on training in production of blow moulded plastic articles 9. Hands on training in production of compression moulded plastic articles		

References

1. C. J. Crawford, *Plastic Engineering*, Pergamon Press, London ,1999.
2. D. H. Morton, *Polymer processing*, Chapman and Hall, London, 1989.
3. George Mathews, *Polymer mixing technology*, Applied Science Publishers, London,1982.
4. Joel Frados (Ed) *Plastic Engineering Hand book*, Van Nostrand Reinhold Company, New York, 1976.

5. Polymer Science and Technology, 2nd Edn. Joel Fried, Prentice Hall of India Ltd.
6. Text Book of Polymer Science, 3rd Edn. Fred W Billmeyer, JR. Wiley
7. Polymer Science, 3rd Edn. VR Gowariker, NV Viswanathan, Jayadev Sreedhar, New Age International Publishers
8. Polymer Chemistry, Dr BK Sharma, Goel Publishing House.

Mapping of COs with PSOs and POs :

	PSO 1	PSO 2	PSO 3	PSO4	PSO5	PSO 6	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO 1	3		1	2	1	2	3		2	2			3
CO 2	3		1	-	-	2	3		2	2			3
CO 3	3	-	2	-	-	3	3		3	2			3
CO 4	3	-	2		-	2	3		1	2			2
CO 5	3		2	-	-	2	2		1	1			3
CO 6	3	-	2		-	2	3		1	2			2
CO 7	3		3			3	3		3	3		2	3

Correlation Levels:

Level	Correlation
-	Nil
1	Slightly / Low
2	Moderate / Medium
3	Substantial / High

Assessment Rubrics:

- Quiz / Assignment/ Quiz/ Discussion / Seminar
- Midterm Exam
- Programming Assignments (20%)
- Final Exam (70%)

Mapping of COs to Assessment Rubrics :

	Internal Exam	Assignment	Project Evaluation	End Semester Examinations
CO 1	✓	✓		✓
CO 2	✓	✓		✓
CO 3	✓	✓		✓
CO 4	✓	✓		✓
CO 5	✓	✓		✓
CO 6	✓	✓		✓
CO7		✓	✓	